

the meaning, integration and dependencies of T shipments in relation to A shipments. The relevant column here is sendungsart on table sendung. i need to understand the business relevance and processes around it.

Date: 2026-03-05

Status: Exploration

Original User Input

Investigate the meaning, integration and dependencies of "T shipments" in relation to "A shipments". The relevant column here is "sendungsart" on table "sendung". Need to understand the business relevance and processes around it.

Summary

Key Finding: The New Dispo system is designed to handle **only 'A' shipments** (Abgangssendung - main/primary shipments). 'T' shipments (Teilsendung - partial/sub-shipments) serve internal TMS routing purposes and are explicitly filtered out at multiple integration points.

- **'A' = Abgangssendung:** Main/primary shipments visible to disposition planning
- **'T' = Teilsendung:** Partial/subsidiary shipments linked to main 'A' shipments, used for multi-leg transport
- **Integration Point:** Only 'A' shipments flow through v_dis_shipment_all view and CDC pipeline to New Dispo
- **Relationship:** 'T' shipments are linked to parent 'A' shipments via the sen_zuord table

Analysis

Business Meaning

'A' = Abgangssendung (Main/Primary Shipment)

- **Definition:** Main or primary shipments in the outbound/dispatch flow
- **Purpose:** The primary shipment that originates from the shipper and is dispatched for delivery
- **Visibility:** These are the operational shipments visible to disposition/pickup planning system
- **Business Process:** Used for standard pickup → main-line transport → delivery flow

'T' = Teilsendung (Partial/Sub-shipment)

- **Definition:** Partial or subsidiary shipments linked to a main 'A' shipment
- **Alternative Names:** "HL-Sendung" (Hauptlaufsendung/main-line), "NL-Sendung" (Nebenlaufsendung/subsidiary)
- **Purpose:** Represent breakdowns or sub-divisions of a main shipment

- **Business Process:** Created when an 'A' shipment is split for operational reasons:
 - Intermediate deliveries at consolidation points
 - Multi-leg transport routing
 - Cross-docking operations
 - Partial deliveries to different locations

Technical Relationship

Sendungsart Column

- **Location:** `sendung.sendungsart`
- **Type:** `character(1)` - single character field
- **Values:**
 - 'A' - Abgangssendung (main shipment)
 - 'T' or 't' - Teilsendung (partial shipment)
 - 'E', 'N', 'F', 'S', 'V', 'W', 'Z' - other shipment types

Parent-Child Linking via `sen_zuord` Table

The relationship between 'A' (parent) and 'T' (child) shipments is managed through the `sen_zuord` table:

- **ref_tix:** Parent shipment TIX (typically 'A' shipment)
- **sen_tix:** Child shipment TIX (typically 'T' shipment)
- **typ:** Relationship type
 - 'H' - HL-Sendung (main-line sub-shipment)
 - 'N' - NL-Sendung (subsidiary sub-shipment)

Traffic Flow (Verkehrsstrom) Determination

'T' shipments are algorithmically marked as "Direktsendung" based on traffic flow:

```
rSen.Direktsendung := case when rSen.Verkehrsstrom in('30 ', '31 ', '32 ', '34 ')
                        then 'T'
                        else 'F'
                        end;
```

Traffic Flows for T shipments:

- '30' - Direct domestic short-haul
- '31' - Domestic long-haul
- '32' - Domestic consolidation
- '34' - International/cross-border

Database Schema

sendung Table (Main Shipment Table)

Location: `Code/tms-alloydb-schema/src/sql/table/sendung.sql:19`

```

CREATE TABLE sendung (
    sendung_tix numeric(22,0) NOT NULL,
    sendungsart character(1),           -- Shipment type: 'A', 'T', 'E', 'N', 'F', etc.
    sendung_n numeric(7,0),             -- Shipment number
    fix_key character(9),               -- Departure service area
    firma numeric(3,0),                 -- Company
    niederlassung numeric(2,0),         -- Branch
    verkehrsstrom character(3),         -- Traffic flow
    direktsendung character(1),         -- Direct shipment flag ('T'/'F')
    ref_sen_tix numeric(22,0),          -- Reference shipment TIX
    ...
);

```

sen_zuord Table (Shipment Association/Linking)

Location: Code/tms-alloydb-schema/src/sql/table/sen_zuord.sql

```

CREATE TABLE sen_zuord (
    ref_tix numeric(22,0) NOT NULL,     -- Parent shipment TIX
    sen_tix numeric(22,0) NOT NULL,     -- Child/associated shipment TIX
    typ character(1) NOT NULL,          -- Association type ('H', 'N', 'S', 'T', 'Z')
    u_version character(1),
    c_time timestamp,
    c_user character(8),
    u_time timestamp,
    u_user character(8),
    lfd_n numeric(5,0)                  -- Sequential number
);

```

Key Constraints

Location: Code/tms-alloydb-schema/src/sql/constraint/pk_uq/sendung_pk_uq.sql:2

```

ALTER TABLE ONLY sendung
    ADD CONSTRAINT sendungc2 UNIQUE (sendung_n, fix_key, sendungsart, firma, niederlassung);

```

The unique constraint includes `sendungsart`, meaning the same shipment number can exist with different `sendungsart` values (e.g., one 'A' and multiple 'T' shipments).

Source Code Evidence

1. Disposition System Integration (New Dispo Backend)

V_DIS_SHIPMENT_ALL View - Only 'A' Shipments

Location: Code/tms-alloydb-schema/src/sql/view/V_DIS_SHIPMENT_ALL.sql

```
create or replace view v_dis_shipment_all as
select
    s.sendung_tix,
    s.sendungsart,
    s.sendung_n,
    ...
from sendung s
where sendungsart = 'A'::bpchar; -- ⚠️ ONLY 'A' shipments visible to New Dispo
```

Impact: The New Dispo batch initialization only receives 'A' shipments via GraphQL queries.

CDC Pipeline Filter

Reference: Cloud Functions filter logic (mentioned in agent findings)

```
shipmentType == 'A' -- Only 'A' shipments published to Pub/Sub
```

Impact: Change events for 'T' shipments do NOT propagate to New Dispo backend.

2. Views Including Both 'A' and 'T' Shipments

V_EMP_SEN_SUM (Recipient Shipment Summary)

Location: Code/tms-alloydb-schema/src/sql/view/V_EMP_SEN_SUM.sql

```
WHERE ((s.verkehrsstrom = ANY (ARRAY['30'::bpchar, '34'::bpchar]))
AND ((s.sendungsart = ANY (ARRAY['t'::bpchar, 'T'::bpchar]))
OR ((s.sendungsart = 'A'::bpchar) AND (sen.isavis(s.status_8) = (1)::numeric))))
```

Business Logic: For specific traffic flows (30, 34), both 'T' and 'A' shipments are included.

V_SEN7, V_SEN4 (General Shipment Lists)

```
WHERE ((s.sendungsart = ANY (ARRAY['A'::bpchar, 'T'::bpchar, 't'::bpchar, 'Z'::bpchar]))
AND (s.verkehrsstrom = ANY (ARRAY['3'::bpchar, '30'::bpchar, '31'::bpchar, '32'::bpchar, '34'::bpchar]))
```

3. Views Explicitly Excluding 'T' Shipments

V_ESB_MEASURINGPOINT5DETAIL (ESB Tracking Measuring Points)

Location: Code/tms-alloydb-schema/src/sql/view/V_ESB_MEASURINGPOINT5DETAIL.sql

```
WHERE ... AND (s.sendungsart <> ALL (ARRAY['F'::bpchar, 'T'::bpchar, 't'::bpchar]))
```

Impact: 'T' shipments do NOT generate ESB measuring point events for external tracking systems.

4. Trigger Logic Handling 'A' Shipments

traiu_sendung Trigger (After Update on sendung)

Location: Code/tms-alloydb-schema/src/sql/trigger/all_trigger_events.sql

```
CREATE TRIGGER traiu_sendung AFTER UPDATE ON sendung
FOR EACH ROW
WHEN (((new.sendungsart = ANY (ARRAY['A'::bpchar, 'E'::bpchar, 'N'::bpchar]))
AND (new.status_erf <> '5'::bpchar) ...))
```

Logic: Main update triggers fire for 'A', 'E', and 'N' shipments, but NOT for 'T' shipments.

5. Backend Entity Mapping

SendungEntity (TMS Bridge)

Location: Code/Disposition-Abstraction-Layer/CALConsult.TMSBridge.API/Data/Entities/Sendung/SendungEntity.cs:22

```
public class SendungEntity
{
    public long SendungTix { get; set; }
    public string? Sendungsart { get; set; } // Maps to sendung.sendungsart
    ...
}
```

GoogleBucketShipmentData (New Dispo Backend)

Location:

Code/Disposition-Backend/CALConsult.Disposition.API/Infrastructure/GooglePubSub/Dtos/GoogleBucketShipmentData.cs:20

```
[JsonProperty("sendungsart")]
public string ShipmentType { get; set; } = null!;
```

Note: Backend receives this field, but upstream filters ensure only 'A' values arrive.

Findings

Critical Integration Points

Component	'A' Shipments	'T' Shipments	Impact
v_dis_shipment_all	✔ Included	✘ Excluded	Only 'A' shipments visible to New Dispo batch init
CDC Pipeline Filter	✔ Published	✘ Filtered out	'T' shipment changes NOT propagated to backend
ESB Measuring Points	✔ Tracked	✘ Not tracked	'T' shipments invisible to external tracking
Disposition Triggers	✔ Triggers fire	✘ Excluded	'A' shipments get full status update processing
sen_zuord Linking	✔ Parent	✔ Child	'T' linked to parent 'A' via ref_tix

Key Business Dependencies

1. Shipment Consolidation
 - Multiple 'T' shipments can be children of a single 'A' shipment
 - Linked via sen_zuord table with typ='H' (HL-Sendung) or typ='N' (NL-Sendung)
2. Pickup Planning (New Dispo)

- Only 'A' shipments are loaded into the disposition system
- Pickup planning operates solely on 'A' shipments
- Driver terminals only see 'A' shipments in pickup lists

3. Tracking & Monitoring

- Only 'A' shipments reach ESB measuring points
- External tracking systems cannot directly track 'T' shipments
- Customer visibility limited to 'A' shipments

4. Status Management

- Dispatch status (status_abf , status_dis) primarily updated for 'A' shipments
- Status triggers focus on 'A', 'E', 'N' types
- 'T' shipments have minimal status transition logic

5. Multi-leg Transport Flow

- 'T' shipments enable multi-stop/multi-leg delivery scenarios
- Represent intermediate routing steps not visible to end customers
- Support consolidation center processing

Processing Differences

'A' Shipments (Abgangssendung)

-  Visible in New Dispo disposition planning
-  CDC events published to backend
-  Full status update triggers
-  ESB measuring point tracking
-  Customer-facing shipment number
-  Pickup planning integration

'T' Shipments (Teilsendung)

-  Hidden from New Dispo disposition planning
-  No CDC events to backend
-  Limited status update triggers
-  No ESB measuring point tracking
-  Linked to parent 'A' shipment via sen_zuord
-  Internal TMS routing purposes only
-  Specific traffic flow handling (30, 31, 32, 34)

Questions/Open Items

Business Requirements Questions

1. **Should 'T' shipments ever be visible in New Dispo disposition planning?**
 - Current state: Only 'A' shipments are visible
 - If yes: Major architectural changes required
2. **What happens when a driver needs to pick up or deliver a 'T' shipment?**
 - Are 'T' shipments always handled at consolidation centers?
 - Do drivers receive separate instructions for 'T' shipments outside of New Dispo?
3. **How does the legacy TMS handle 'T' shipments in dispatch planning?**
 - Is there a separate disposition process for 'T' shipments?
 - Are they automatically assigned based on parent 'A' shipment routing?

4. Customer Visibility:

- Can customers track 'T' shipments independently?
- Or do they only see the parent 'A' shipment status?

5. Multi-leg Routing:

- How are 'T' shipments created? Manual or automatic?
- What triggers the creation of a 'T' shipment from an 'A' shipment?

Technical Implementation Questions

6. If 'T' shipment integration is required, what changes are needed?

- Modify `v_dis_shipment_all` to include 'T' shipments?
- Update CDC filter to propagate 'T' shipment events?
- Add backend domain logic to handle parent-child relationships?
- UI changes to display linked shipments?

7. Leg/Lot Generation:

- Should 'T' shipments generate separate legs?
- Or should legs reference the parent 'A' shipment?

8. Status Synchronization:

- Should status updates on 'T' shipments affect parent 'A' shipment?
- Vice versa?

9. Route Calculation:

- Are 'T' shipments independently routed?
- Or do they inherit routing from parent 'A' shipment?

10. Data Integrity:

- Can a 'T' shipment exist without a parent 'A' shipment?
- What validation rules apply to `sen_zuord` relationships?

Related Files

Database Schema (TMS AlloyDB)

- **Main Table:** `Code/tms-alloydb-schema/src/sql/table/sendung.sql` (line 19: `sendungsart` column)
- **Linking Table:** `Code/tms-alloydb-schema/src/sql/table/sen_zuord.sql`
- **Constraints:** `Code/tms-alloydb-schema/src/sql/constraint/pk_uq/sendung_pk_uq.sql` (line 2: unique constraint)

Views (Critical for Integration)

- **New Dispo Integration:** `Code/tms-alloydb-schema/src/sql/view/V_DIS_SHIPMENT_ALL.sql` (filters for 'A' only)
- **Recipient Summary:** `Code/tms-alloydb-schema/src/sql/view/V_EMP_SEN_SUM.sql` (includes both 'A' and 'T')
- **Shipment Lists:** `Code/tms-alloydb-schema/src/sql/view/V_SEN7.sql` , `V_SEN4.sql` (includes 'A', 'T', 'Z')
- **ESB Tracking:** `Code/tms-alloydb-schema/src/sql/view/V_ESB_MEASURINGPOINT5DETAIL.sql` (excludes 'T')

Triggers

- **Update Trigger:** `Code/tms-alloydb-schema/src/sql/trigger/all_trigger_events.sql` (`traiu_sendung` - handles 'A', 'E', 'N')
- **Post-View Functions:** `Code/tms-alloydb-schema/src/sql/trigger/all_trigger_functions_post_views.sql` (`Direktsendung` logic)

Stored Procedures/Packages

- **SEN Package:** Code/tms-alloydb-schema/src/sql/package/SEN.sql (lines 11092, 13817: 'E','N' filtering)
- **PPST Package:** Code/tms-alloydb-schema/src/sql/package/PPST.sql (HL-/NL-Sendung comments and logic)
- **PFRK Package:** Code/tms-alloydb-schema/src/sql/package/PFRK.sql (lines 1132, 3207: 'A' filtering)
- **DISP_MDE_AH:** Code/tms-alloydb-schema/src/sql/package/DISP_MDE_AH.sql (line 178: 'A','N' filtering)

Backend Code (New Dispo)

- **TMS Bridge Entity:**
Code/Disposition-Abstraction-Layer/CALConsult.TMSBridge.API/Data/Entities/Sendung/SendungEntity.cs (line 22: Sendungsart property)
- **CDC DTO:**
Code/Disposition-Backend/CALConsult.Disposition.API/Infrastructure/GooglePubSub/Dtos/GoogleBucketShipmentData.cs (line 20: ShipmentType mapping)

Documentation

- **Architecture Docs:** Check 08_Documentation/ folder for any shipment data flow documentation
- **Related Exploration:** 02_Explorations/2026-02-25_User_Story_103821_OMS_Sendung_Quell_K_analysis/ (quell_k field related)

Related User Stories/Tasks

Potentially Impacted Areas

1. **User Story: OMS Sendung Quell_K analysis** (2026-02-25 exploration)
 - Related exploration on sending table fields
 - May have implications for 'T' shipment handling
2. **Pickup Planning / Batch Initialization**
 - Current: Only loads 'A' shipments via v_dis_shipment_all
 - Impact: Any requirements to plan 'T' shipments would require view changes
3. **CDC Pipeline / Change Notifications**
 - Current: Filters for shipmentType == 'A' in Cloud Functions
 - Impact: Adding 'T' shipment events would require filter modification
4. **Route Calculation**
 - Question: Should 'T' shipments have independent routes?
 - Impact: May need separate route calculation logic for multi-leg scenarios
5. **Driver Terminal / Mobile App**
 - Current: Only displays 'A' shipments
 - Impact: If 'T' shipments need driver visibility, UI changes required
6. **Customer Tracking / Portal**
 - Question: Should customers see 'T' shipment details?
 - Impact: May need API changes to expose linked shipments

Future Considerations

- **Multi-leg Transport Enhancement:** If New Dispo needs to support multi-leg routing, 'T' shipment integration becomes critical
- **Consolidation Center Integration:** 'T' shipments are key for consolidation/cross-dock workflows
- **International Shipments:** Traffic flows 34 (international) heavily use 'T' shipments for border crossing/customs handling

Deep Dive: T and A Shipments in Legacy TMS Context

This section explores how T and A shipments work within the legacy TMS database, independent of New Dispo integration.

Complete Shipment Type Classification

The `sendungsart` field defines a comprehensive shipment type hierarchy:

Type	German Name	English Translation	Purpose
'S'	Transportauftrag (TA)	Transport Order	Master shipment aggregating other shipments
'A'	Abgangssendung	Outbound Shipment	Regular shipment with direct delivery
'T'	Teilsendung	Partial/Split Shipment	Created from splitting A or E shipments
'Z'	ZUSA	Consolidation Shipment	Aggregation/consolidation of multiple shipments
'E'	Einzelsendung	Single Shipment	Standalone single-piece shipment
'N'	Nachlieferung	Follow-up Shipment	Subsequent delivery for same customer
'F'	Fraktion	Fraction/Sub-unit	Sub-unit of larger shipment
'U'	Unknown/Conversion	-	Conversion type

Key Insight: T and A shipments are part of a larger hierarchy where 'S' (Transport Orders) serve as parent containers, and 'Z' (ZUSA) shipments provide consolidation functionality.

Shipment Hierarchy Model

```
S (Transportauftrag/TA) - Transport Order
├─ A (Abgangssendung) - Direct outbound shipment
│   └─ [Optional] T (Teilsendung) - If split via loading list
│       └─ [Optional] Z (ZUSA) - Consolidation
├─ E (Einzelsendung) - Single-piece shipment
│   └─ [Optional] T (Teilsendung) - If split
├─ N (Nachlieferung) - Follow-up delivery
└─ F (Fraktion) - Fraction/sub-unit
```

Linking Mechanism: All relationships managed through `sen_zuord` table:

- `typ = 'S'` : Links S shipment to child shipments (A/T/E/N)
- `typ = 'Z'` : Links Z consolidation shipment to consolidated shipments

Source: `Code/tms-alloydb-schema/src/sql/view/v_ta_sen7.sql`

sen_zuord Relationship Types

```
CREATE TABLE sen_zuord (  
    ref_tix numeric(22,0) NOT NULL, -- Parent shipment TIX  
    sen_tix numeric(22,0) NOT NULL, -- Child shipment TIX  
    typ character(1) NOT NULL,      -- Relationship type  
    lfd_n numeric(5,0)              -- Sequence number for ordering  
);
```

Relationship Types (typ field):

- 'S' - Standard TA linkage: Parent S (Transport Order) → Child shipment (A/T/E/N)
- 'Z' - ZUSA linkage: Z (Consolidation) → Consolidated shipments
- 'H' - HL-Sendung: Main-line sub-shipment (referenced in comments)
- 'N' - NL-Sendung: Secondary/subsidiary shipment (referenced in comments)

T Shipment Creation Workflow

When Are T Shipments Created?

Trigger 1: Loading List Assignment

```
-- From V_TA_ZUSA_SEN.sql  
WHERE s.sendungsart = 'T'  
      AND s.ladelist_tix IS NOT NULL
```

When an A shipment gets assigned to a loading list (ladelist_tix):

1. Original A shipment remains as master record
2. T shipments created for each piece/colli on the loading list
3. Each T shipment references the same ladelist_tix
4. T shipments linked to parent S via sen_zuord(typ='S')

Trigger 2: Multi-leg Routing Requirements

Traffic flows 30, 31, 32, 34 automatically mark shipments for splitting:

```
-- From all_trigger_functions_post_views.sql  
rSen.Direktsendung := case  
    when rSen.Verkehrsstrom in('30 ', '31 ', '32 ', '34 ')  
    then 'T'  
    else 'F'  
end;
```

When direktsendung = 'T', the shipment is eligible for T-type processing.

Trigger 3: Piece-Based Operations

From LSTGEN.sql (Loading List Generation):

```
(i_lst_art = 'N' AND (i_sendung.sendungsart = 'A' OR i_sendung.sendungsart = 'F'))  
(i_lst_art = 'VH' AND (i_sendung.sendungsart = 'A' OR i_sendung.sendungsart = 'F'))  
(i_lst_art = 'H' AND (i_sendung.sendungsart = 'A' OR i_sendung.sendungsart = 'F'))
```

Only 'A' and 'F' shipments can generate loading lists, which then create 'T' shipments.

Document Assignment: Bordero vs Rollkart vs Ladelist

A (Abgangssendung) Document Flow:

```
-- Key fields in sendung table
bordero_tix numeric(22,0),      -- Bordero (waybill) TIX
bordero_n  numeric(15,0),      -- Bordero number
bordero_pos numeric(3,0),      -- Position on bordero
bordero_e  timestamp,          -- Bordero entry time

rollkart_tix numeric(22,0),     -- Rollkart (roll cart) TIX
rollk_n    numeric(15,0),      -- Rollkart number
rollk_pos  numeric(3,0),       -- Position on rollkart

ladelist_tix numeric(22,0),     -- Loading list TIX
ladeliste_n numeric(15,0),     -- Loading list number
ladeliste_pos numeric(3,0)     -- Position on loading list
```

Document Priority:

1. **Bordero** - Primary shipping document for A shipments
 - Used for standard direct deliveries
 - Contains routing and delivery instructions
 - Drives status transitions
2. **Rollkart** - For smaller consolidated shipments
 - Multiple shipments on same physical cart
 - Often used for local/regional distribution
3. **Ladelist** - Triggers T shipment creation
 - Multi-piece shipments requiring piece-level tracking
 - Creates one T shipment per piece or logical group
 - Enables independent routing per piece

T (Teilsendung) Document Flow:

```
-- From V_TA_ZUSA_SEN.sql
WHERE (s.sendungsart = 'T' AND s.ladelist_tix IS NOT NULL)
```

T Shipments:

- **Always** have ladelist_tix populated (by definition created from loading lists)
- May **inherit** bordero_tix from parent A shipment
- Can have **independent** rollkart_tix for physical routing
- Share same ladeliste_n with sibling T shipments

Status Field Workflow Differences

Status Fields Overview:

status_erf	character(1)	-- Erfassung (Capture/Entry) status
status_dis	character(1)	-- Disposition status
status_abf	character(1)	-- Abfertigung (Dispatch) status
status_frb	character(1)	-- Frachtberechnung (Freight calculation)
status_zus	character(1)	-- Zusatz (Supplementary)
status_mod	character(1)	-- Modifikation (Modification)
status_rue	character(1)	-- Rückmeldung (Feedback/Response)
status_sta	character(1)	-- Status aggregation
status_hst	numeric(3,0)	-- HST (History) status pointer
status_8	character(1)	-- AVIS flag (advance notice)

A Shipment Status Processing:

Trigger Logic:

```
-- From all_trigger_events.sql
CREATE TRIGGER trau_sendung AFTER UPDATE ON sendung
WHEN (
    -- Case 1: Status and address changes for A/E/N shipments
    (new.sendungsart = ANY (ARRAY['A'::bpchar, 'E'::bpchar, 'N'::bpchar]))
    AND (new.status_erf <> '5'::bpchar)

    OR

    -- Case 2: Dispatch status change for A shipments only
    (((COALESCE(new.status_abf, '*') <> COALESCE(old.status_abf, '*'))
    AND (new.sendungsart = 'A'::bpchar)
    AND (new.status_rue IS NULL)
    AND ((new.status_abf = 'A'::bpchar) OR (old.status_abf = 'A'::bpchar)))
)
```

Key Points:

- **status_abf** changes trigger workflow ONLY for 'A' shipments
- When status_abf = 'A' (dispatched), special processing occurs
- Requires status_rue IS NULL (no feedback response yet)
- 'E' and 'N' shipments get address/routing change triggers
- **'T' shipments excluded from main trigger logic**

T Shipment Status Processing:

From view logic in V_TA_SEN.sql :

```

-- Mode-dependent T shipment visibility
WHEN (
    (COALESCE(pta.getmodus(), 1) <> 1) -- Non-standard mode
    AND (
        (s.sendungsart = ANY (ARRAY['t'::bpchar, 'T'::bpchar]))
        OR ((s.sendungsart = 'A'::bpchar) AND (sen.isavis(s.status_8) = 1))
    )
)
OR
(
    (COALESCE(pta.getmodus(), 1) = 1) -- Standard mode
    AND (s.sendungsart <> ALL (ARRAY['T'::bpchar, 't'::bpchar]))
)

```

Interpretation:

- **Standard mode (modus=1):** T shipments hidden from TA views
- **Alternative modes:** T shipments visible alongside A shipments with AVIS flag
- **AVIS (status_8):** When set, A shipments treated similarly to T shipments in reporting

Why the difference?

- T shipments represent internal operational splits
- Customers and standard workflows see only A shipments
- Warehouse/logistics operations need T visibility in special modes

ZUSA (Consolidation) Shipment Integration

The Z Shipment Role:

Purpose: Consolidate multiple A or T shipments for combined transport

View: V_TA_ZUSA_SEN

```

-- ZUSA relationships
WHERE (
    -- Case 1: T shipments with loading lists linked to other T
    (s.sendungsart = 'T' AND s.ladelist_tix IS NOT NULL AND z.sendungsart = 'T')

    OR

    -- Case 2: A/U shipments linked to Z consolidation
    ((s.sendungsart = 'A' OR s.sendungsart = 'U') AND z.sendungsart = 'Z')
)
AND z.status_3 = 'Z' -- ZUSA marker in status field

```

Consolidation Pattern:

```

S (Transport Order)
├─ Z (ZUSA Consolidation)
│   ├─ A (Shipment 1)
│   ├─ A (Shipment 2)
│   └─ T (Shipment 3) ← from split A shipment

```

Business Use Case:

- Multiple customer shipments going to same geographic region
- Combined into single Z shipment for efficient trunk transport
- De-consolidated at destination hub
- Individual A/T shipments delivered separately at final mile

Multi-leg Routing with T Shipments

Leg Structure:

T shipments enable multi-hop routing through resource history tracking:

From V_TA_SEN.sql:

```
t.res_hst_tix,          -- Resource/Hardware Stop TIX  
t.typ = 3 AND t.art = 100 -- Tour point type indicator
```

Multi-leg Pattern:

```
A (Origin Shipment) → [Split at Hub 1]  
  ↳ T1 (Leg: Hub 1 → Hub 2) → res_hst_tix = Stop1  
  |   ↳ T1.1 (Leg: Hub 2 → Final Dest) → res_hst_tix = Stop2  
  |  
  ↳ T2 (Leg: Hub 1 → Hub 3) → res_hst_tix = Stop3  
    ↳ T2.1 (Leg: Hub 3 → Final Dest) → res_hst_tix = Stop4
```

Status Progression:

- Each leg has independent status_dis , status_abf tracking
- Parent A shipment status aggregates child T statuses
- res_hst_tix links to physical stop/location history

Dispatch Logic from DISP_SPEICH.sql:

```
IF (v_disp_info = 'LK'  
    OR (v_disp_info = 'LT' AND v_sendungsart = 'A')  
    OR (v_disp_info = 'LM' AND v_sendungsart != 'A'))
```

Dispatch Type Codes:

- **LK** - Any shipment type (standard)
- **LT** - Only for 'A' shipments (direct transport)
- **LM** - Only for non-'A' shipments (T, E, N - multi-leg or special)

Key TMS Reporting Views

V_TA_SEN7 - Complete Shipment Hierarchy

Structure:

```

-- Shows: S (TA) → Z (ZUSA) → Shipments (A/T/E/N)
SELECT
    t.sendung_tix AS ta_tix,      -- Transport Order TIX
    z.sendung_tix AS zusa_tix,   -- ZUSA TIX (or direct child)
    s.sendung_tix AS sen_tix     -- Final shipment TIX
FROM sendung t, sen_zuord tz, sendung z, sen_zuord zs, sendung s
WHERE t.sendungsart = 'S'
    AND z.sendungsart IN ('T', 'Z')
    AND s.sendungsart IN ('A', 'U', 'T', 'E', 'N', 'H')

```

Source: Code/tms-alloydb-schema/src/sql/view/v_ta_sen7.sql

Use Case: Complete visibility of entire transport order hierarchy for operations planning

V_TA_ZUSA_SEN - Consolidation Analysis

Structure:

```

-- Shows: TA → ZUSA → A/T shipments with consolidation
WHERE (
    (s.sendungsart = 'T' AND s.ladelist_tix IS NOT NULL AND z.sendungsart = 'T')
    OR
    ((s.sendungsart = 'A' OR s.sendungsart = 'U') AND z.sendungsart = 'Z')
)
AND z.status_3 = 'Z'

```

Source: Code/tms-alloydb-schema/src/sql/view/V_TA_ZUSA_SEN.sql

Use Case: Analyze consolidation efficiency, track consolidated shipment groups

V_EMP_SEN_SUM - Recipient-Based Aggregation

Structure:

```

-- Aggregates by recipient, includes T and A with AVIS
WHERE (
    (s.verkehrsstrom = ANY (ARRAY['30'::bpchar, '34'::bpchar]))
    AND (
        (s.sendungsart = ANY (ARRAY['t'::bpchar, 'T'::bpchar]))
        OR ((s.sendungsart = 'A'::bpchar) AND (sen.isavis(s.status_8) = 1))
    )
)
GROUP BY empf_n, empf_i, empf_name1, emp_erm_rel

```

Source: Code/tms-alloydb-schema/src/sql/view/V_EMP_SEN_SUM.sql

Use Case: Customer service view showing all shipments (including T pieces) for recipient

V_REL_SEN_SUM - Service Area Aggregation

Structure:

```
-- Aggregates by service area (relation)
WHERE sendungsart IN ('T', 't')
      OR (sendungsart = 'A' AND sen.isavis(status_8) = 1)
GROUP BY emp_erm_rel, leistungsdatum
```

Use Case: Regional operations planning by service area

Status Aggregation and Reporting

AVIS Flag (status_8) Special Handling:

The `status_8` field serves as an AVIS (advance notice) flag:

```
-- From sen.isavis() function logic in views
sen.isavis(s.status_8) = 1 -- AVIS is active
```

When AVIS is set:

- A shipments treated like T shipments in some reports
- Indicates customer requested special notification
- Changes visibility in recipient and service area views
- Aggregated separately in summary reports

Business Meaning:

- Customer wants advance notice of delivery
- Shipment requires special handling/communication
- Often used for high-value or time-critical shipments

German Business Terminology Reference

German Term	Abbreviation	English	TMS Field/Context
Transportauftrag	TA	Transport Order	sendungsart = 'S'
Abgangssendung	-	Outbound Shipment	sendungsart = 'A'
Teilsendung	-	Partial Shipment	sendungsart = 'T'
Zusammenfassung	ZUSA	Consolidation	sendungsart = 'Z', status_3 = 'Z'
Einzelsendung	-	Single Shipment	sendungsart = 'E'
Nachlieferung	-	Follow-up Delivery	sendungsart = 'N'
Hauptlaufsendung	HL	Main-line Shipment	sen_zuord.typ = 'H'
Nebenlaufsendung	NL	Secondary Shipment	sen_zuord.typ = 'N'
Bordero	-	Waybill/Bill of Lading	bordero_tix, bordero_n
Rollkarte	-	Roll Cart	rollkart_tix, rollk_n
Ladeliste	-	Loading List	ladelist_tix, ladeliste_n
Erfassung	ERF	Data Capture/Entry	status_erf

German Term	Abbreviation	English	TMS Field/Context
Disposition	DIS	Planning/Assignment	status_dis
Abfertigung	ABF	Dispatch/Handling	status_abf
Rückmeldung	RUE	Feedback/Response	status_rue
Avis	-	Advance Notice	status_8
Verkehrsstrom	-	Traffic Flow	verkehrsstrom field
Relation	REL	Service Area	relation, emp_erm_rel

Complete T/A Lifecycle Workflow

1. ORDER ENTRY

- ↳ Create S (Transport Order) – sendungsart = 'S'
 - ↳ Order contains one or more planned shipments
- ↳ Links via sen_zuord to child shipments

2. SHIPMENT CREATION

- ↳ Create A (Abgangssendung) – sendungsart = 'A'
 - ↳ Linked to parent S via sen_zuord(typ='S')
 - ↳ status_erf set (data captured)
 - ↳ Address and routing info populated
- ↳ Assigned to bordero (standard workflow)

3. PLANNING DECISION POINT

Path A: DIRECT DELIVERY (no splitting)

- ↳ A shipment proceeds as-is
 - ↳ status_dis = 'L' (planned to tour)
 - ↳ Assigned to driver/tour
 - ↳ status_abf = 'A' (dispatched) triggers workflow
- ↳ Delivered as single unit

Path B: MULTI-PIECE / MULTI-LEG (splitting)

- ↳ Loading list assignment
 - ↳ ladelist_tix populated on A shipment
 - ↳ Generate T shipments for each piece/group
 - ↳ Each T has same ladelist_tix
 - ↳ Each T linked via sen_zuord(typ='S') to parent S
 - ↳ T shipments can have independent routing
 - ↳ T shipments tracked separately through legs
- ↳ Original A remains as master record

4. CONSOLIDATION (optional)

- ↳ Create Z (ZUSA) consolidation
 - ↳ Multiple A or T shipments combined
 - ↳ Linked via sen_zuord(typ='Z')
 - ↳ status_3 = 'Z' marks ZUSA type
 - ↳ Combined transport to shared destination
- ↳ De-consolidated at destination hub

5. DISPATCH & TRACKING

For A shipments:

- ↳ status_abf updates trigger main workflow
- ↳ ESB measuring points generated
- ↳ Customer tracking available
- ↳ Standard status progression

For T shipments:

- ↳ Independent status per leg/piece
- ↳ No ESB measuring points (internal only)
- ↳ Not visible to customer directly
- ↳ Aggregated status rolled up to parent A
- ↳ Special mode required for TMS visibility

6. DELIVERY COMPLETION

- ↳ All T pieces completed → A status updated
 - ↳ status_rue processed (feedback)
 - ↳ Customer sees A shipment as delivered
- ↳ T shipment details archived

Critical Files for T/A Logic

Core Tables:

- Code/tms-alloydb-schema/src/sql/table/sendung.sql (line 19: sendungsart)
- Code/tms-alloydb-schema/src/sql/table/sen_zuord.sql (relationship table)

Hierarchy Views:

- Code/tms-alloydb-schema/src/sql/view/v_ta_sen7.sql (complete hierarchy)
- Code/tms-alloydb-schema/src/sql/view/V_TA_ZUSA_SEN.sql (consolidation)
- Code/tms-alloydb-schema/src/sql/view/V_TA_SEN.sql (TA → shipment detail)

Summary Views:

- Code/tms-alloydb-schema/src/sql/view/V_EMP_SEN_SUM.sql (recipient aggregation)
- Code/tms-alloydb-schema/src/sql/view/V_REL_SEN_SUM.sql (service area aggregation)

Processing Logic:

- Code/tms-alloydb-schema/src/sql/package/PTA.sql (TA processing)
- Code/tms-alloydb-schema/src/sql/package/LSTGEN.sql (loading list generation)
- Code/tms-alloydb-schema/src/sql/package/DISP_SPEICH.sql (dispatch storage)
- Code/tms-alloydb-schema/src/sql/trigger/all_trigger_events.sql (status triggers)

Key Insights for Architecture Decisions

1. T Shipments Are Not Independent Entities

- Always children of S (Transport Order) shipments
- Created as byproduct of loading list assignment
- Cannot exist without parent S linkage via sen_zuord

2. A Shipments Are Customer-Facing

- Primary business transaction record
- Visible in customer tracking systems
- Generate ESB measuring point events
- Trigger main workflow status transitions

3. T Shipments Enable Internal Operations

- Support multi-leg routing without customer complexity
- Enable piece-level tracking for large shipments
- Allow consolidation/de-consolidation flexibility
- Keep customer view simple while supporting complex logistics

4. Status Synchronization Is One-Way

- T shipment status → rolls up to parent A shipment
- A shipment status → does NOT automatically update T shipments
- Each T shipment has independent operational status
- Customer sees only aggregated A shipment status

5. Integration Implications for New Dispo

- Current filtering (A only) maintains customer-facing view
- Adding T shipments would expose internal operations
- Would require parent-child relationship handling in domain model
- May create confusion if drivers see internal split pieces
- Consider whether multi-leg planning is needed vs. single-leg focus